

Physics 216: Electronics

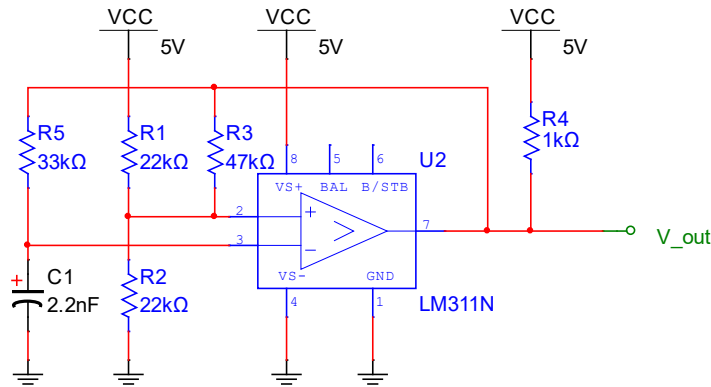
“M” problems

Part C: Capacitors, Inductors, and AC circuits

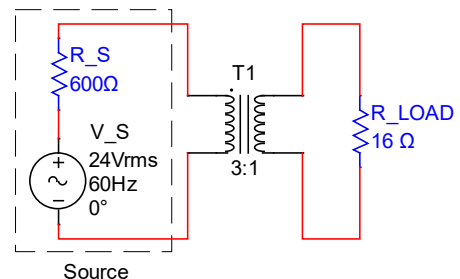
Answers to problems are at the end of this document.

MC1. A 200 nF capacitor is charged by a 5 V supply through a 2 k Ω resistor, starting at zero volts. (a) Graph both the current I_C through the capacitor and the voltage V_C across the capacitor as functions of time. (b) At what time does the capacitor voltage reach 90% of the supply voltage? (c) What is the current through the resistor at the time you calculated in (b)?

MC2. Consider the circuit shown below. (a) Calculate the two possible values of the input terminal V_+ , (the upper and lower threshold voltages) depending on the value of V_{out} . You can assume $R_4 \approx 0$. (b) Draw graphs of voltage vs. time for V_{out} , V_+ , and V_- , showing how they are synchronized. (c) Calculate the time Δt_{rise} for V_- to rise from the lower threshold voltage to the higher threshold voltage. (d) Calculate the time Δt_{fall} for V_- to fall from the higher threshold voltage to the lower threshold voltage. (e) Find the output frequency of the oscillator. *Note: on an exam, you might be given this drawing and asked to find the frequency, without being asked all the helpful leading questions (a) through (d).*

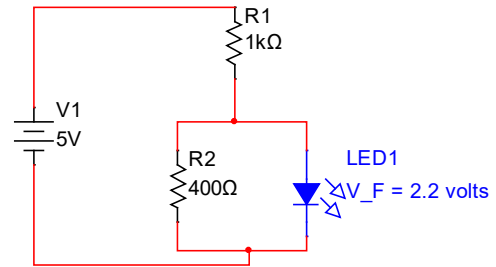


MC3. The circuit to the right shows a voltage source with an internal impedance connected to a load via a transformer. (a) What is the resistance of the apparent load “seen” by the source? (b) What is the voltage across the transformer’s primary coil? (c) What is the average power delivered to the load resistor, as shown? (d) If you could replace the transformer, what turns ratio would you choose to deliver the maximum power to the load resistor? (e) What is the maximum power you could deliver to the load?



MC4. Our goal in this problem is to analyze the circuit below, finding the voltage and current of each component. We can't tell ahead of time if there is current through the diode or not, so we'll try both possibilities.

Assume for parts (a) through (d) that there *is* a forward current in the diode.

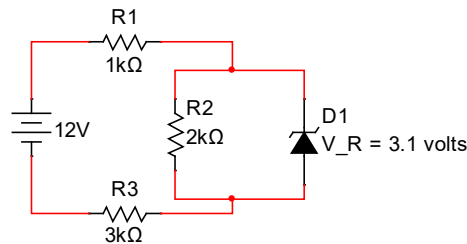


- Assuming there's a forward current in the diode, what is the voltage V_D across the diode?
- What are V_{R1} and V_{R2} ?
- What are I_{R1} and I_{R2} ?
- What is it about your last three answers that is inconsistent, thus demonstrating that our assumption in part (a) must be bogus?

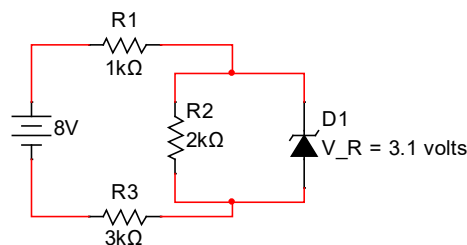
Now assume that there is NO current in the diode.

- Assuming no current in the diode, does the diode behave like an open or like a short?
- Find V_{R1} , V_{R2} , and the current in the circuit.
- Are the voltages you found in part f consistent with your assumption in part e?

MC5. This problem is a lot like the previous one, except that you won't be led through it step by step. For the circuit below, find the voltages and currents for all resistors and the diode.



MC6. In this problem, the battery from the previous circuit has been replaced. Find the voltages and currents for all resistors and the diode.

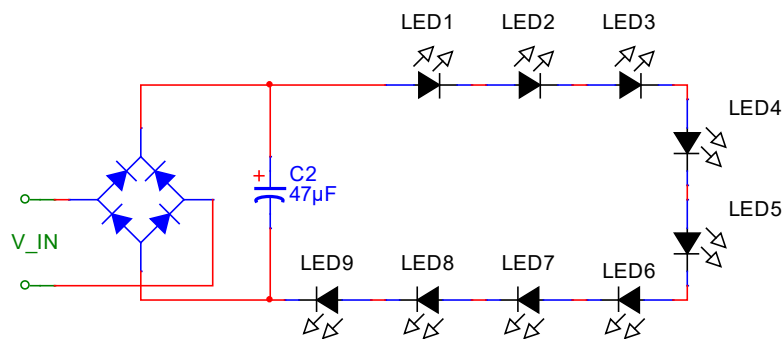


MC7. Do a little Googling for the 1N4001 diode you've been using in the lab, and find (a) its reverse breakdown voltage V_R and (b) the maximum steady current you can pass through it in the forward direction, above which it presumably starts to smoke. Cite your source or sources. (I know I've already posted the datasheet for the 1N4001 online, but see if you can find the information on your own.)

MC8. Design two clipper circuits (series or shunt, your choice) for each of the following tasks:

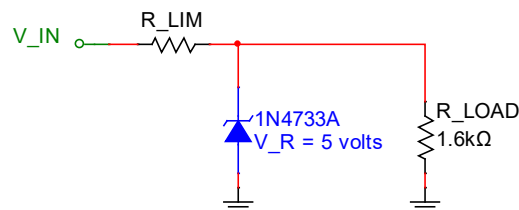
(a) The first circuit should accept an input of $-10\text{ V} < V_{\text{in}} < +10\text{ V}$ and produce an output that is generally positive, clipping the negative part. For this, you don't particularly care whether the exact threshold for clipping is 0.0 or $\pm 0.7\text{ V}$, but you do care that for V_{in} above the threshold, V_{in} and V_{out} should differ by no more than 0.5% . The load for this application will be about $R_{\text{load}} = 50\text{ k}\Omega$. (b) The second circuit should also accept an input of $-10\text{ V} < V_{\text{in}} < +10\text{ V}$, but produce an output that is generally negative, clipping the positive part. You don't care about the precision of the output much: either the exact clipping threshold or the exact ratio of $V_{\text{out}}/V_{\text{in}}$. But this circuit will be used for $R_{\text{load}} = 50\text{ }\Omega$, drawing up to 200 mA . Make sure that your circuit can handle this current.

MC9. The diagram below shows part of a circuit inside of an actual screw-in LED light bulb. The AC voltage input V_{in} has already been reduced from 120 volts RMS to something smaller. With the bulb turned on, the voltage across the capacitor is measured to have an *average* (DC) level of 28.7 volts and a ripple voltage of 3.4 volts . (a) Rewrite your expression for ripple voltage you derived in the lab in terms of the maximum current I_{max} instead of V_{max} . (b) For small ripple voltages, $I_{\text{max}} \approx I_{\text{avg}}$. In fact, in your derivation, you used $\Delta Q = I\Delta t$. Which I is technically more accurate there, I_{avg} or I_{max} ? (c) Based on the measurements given, find a numerical value for the average current through the LEDs. (d) Find the total power dissipated by the LEDs (some of which, presumably, is in the form of visible light). (e) Find the peak input voltage V_{in} required to make the bulb work as measured above.

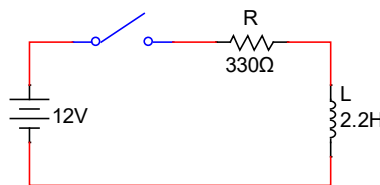


MC10. This problem will step you through process of designing a 5-volt DC power supply that can supply up to 1 amp of current, with a ripple voltage of less than 2 mV . You will use a full wave bridge rectifier and an MC7805 regulator, which has an output voltage of 5 V , a dropout voltage of 2 V , and a minimum ripple rejection of 62 dB . (a) What is the minimum instantaneous input voltage for the regulator? (b) What is the minimum ripple voltage at the input of the regulator to keep the output ripple from exceeding 2 mV ? (c) What value of capacitor is required to maintain that ripple voltage? (d) What is the peak output voltage required from the transformer? (e) Assume the AC line is nominally 120 volts RMS , but might drop as low as 105 volts . What turns ratio transformer is required to ensure a stable 5 V output?

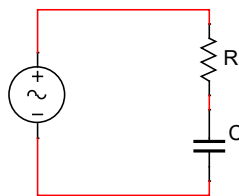
MC11. In the circuit below, a Zener shunt clipper is being used to regulate the voltage across $R_{\text{load}} = 1.6 \text{ k}\Omega$. The input voltage V_{in} varies sinusoidally between 7.5 and 9.0 volts. (a) What value of R_{lim} is required for the circuit to function properly, and is that a minimum or a maximum value for R_{lim} ? (b) Assuming the value of R_{lim} from the last part is chosen, what is the average current through the diode? (c) What is the maximum instantaneous power in both R_{lim} and the diode?



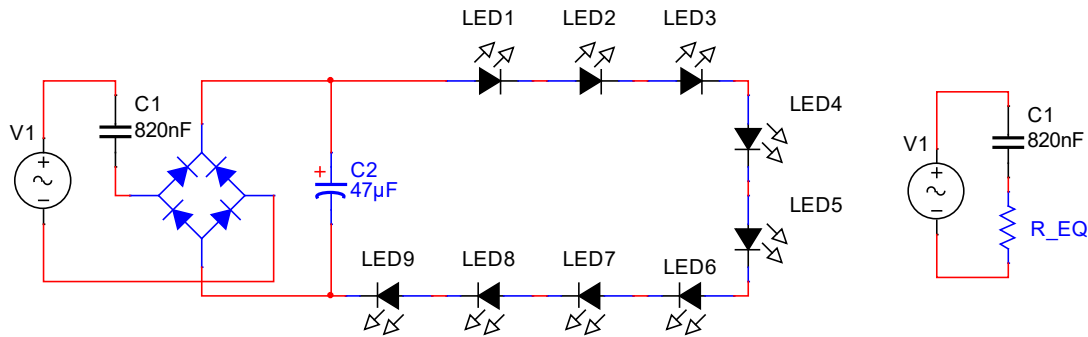
MC12. In the circuit below, the switch is closed at time $t = 0$. (a) Find the current I_L through the inductor at $t = 10 \text{ msec}$. (b) Is the current through the resistor I_R the same as through the inductor at that time? (c) Find the voltage across the inductor V_L at that time. *Hint: you should have found in Lab 13 that the time constant for an "LR" circuit is given by $\tau = L/R$.*



MC13. This problem is analogous to the example we did in class, and for which I have posted an example problem on Blackboard. Consider the circuit below, with a sinusoidal voltage source $V_S(t) = V_0 e^{i\omega t}$. (a) Find the magnitude and phase of the current $I(t)$ in the circuit. (Both are functions of ω .) (b) Find the magnitude and phase of the voltage V_C across the capacitor. (c) Find the "cutoff frequency" f_c at which the ratio of $|V_C|/V_0 = 1/\sqrt{2}$, or about 0.707. (d) At this frequency, what is $|V_C|/V_0$ expressed in decibels? (e) Find expressions for $|V_C|/V_0$ in the limit of low frequency and in the limit of high frequency. (f) Draw vectors on the complex plane for V_S , V_R , V_C , and I .

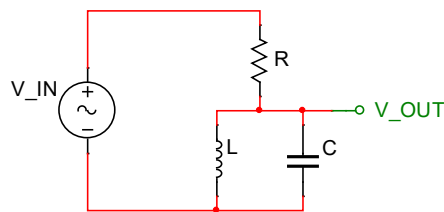


MC14. (True story!) A while ago I found some LED light bulbs for sale online that looked like a real bargain. They were just the intensity I needed and they were *super cheap*, at a time when most LED bulbs were a lot pricier. Unfortunately, after buying a bunch of them, I realized I had made a mistake: they were from China and were designed to work with 220 VAC, instead of 120 VAC like in the US. D’oh!!! I opened one up and found the circuit below, on the left.

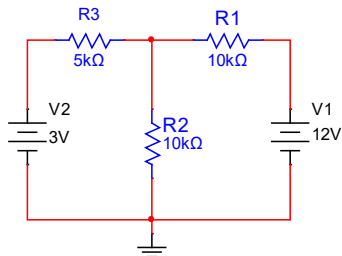


The bulb uses what’s called a “drop cap” design, in which the capacitor C_1 acts as a voltage divider with the rest of the circuit, yielding a reduced (“dropped”) voltage across the rectifier. For this problem, we’ll make the gross simplification that the entire rest of the circuit acts like just an equivalent resistor R_{EQ} , as shown in the circuit on the right. That way, you can treat the circuit using complex numbers, directly using your results from problem MC13. (a) Given an input of $V_1 = 220$ V RMS ($V_{pk} = 311.1$ V), and a *peak* voltage of 31.8 V across the rectifier (or across R_{EQ}), what is the apparent value of R_{EQ} ? (b) For the values in the previous part, what is the power dissipated in C_1 ? (c) Now suppose that the bulb is to be used with an input voltage of $V_1 = 120$ V RMS. What new value of capacitor C_1 is needed to maintain a peak voltage of 31.8 V across the rectifier? (d) In fact, I bought some larger capacitors, soldered them into the bulbs, and those LED bulbs remain in my house to this day, where they work just great! Isn’t Matt clever for figuring this out?

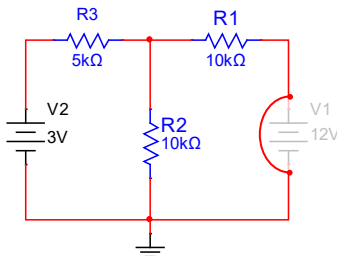
MC15. For the circuit below, $V_{in} = V_0 e^{i\omega t}$. (a) Use the same techniques that you used in problem MC13 to derive algebraic expressions for the magnitude and phase of V_{out} . (b) What is the very special value of ω (in terms of L and C) for which V_{out} is maximized? (c) What are the magnitude and phase of V_{out}/V_{in} for the three cases of (i) very low frequency, (ii) very high frequency, and (iii) $\omega = 1/\sqrt{LC}$?



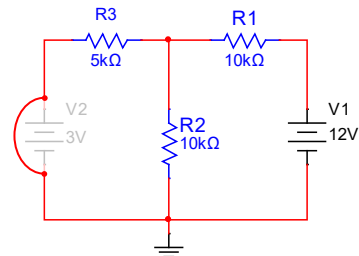
MC16. In this problem you will analyze the circuit on the left, below, using the *principle of superposition*, which is also described in chapter 7 of Paynter. (a) Imagine replacing the source V_2 with a short, and calculate the voltage across R_2 . (b) With source V_2 restored, replace the source V_1 with a short and calculate the voltage across R_2 . (c) The actual voltage across R_2 for the circuit as drawn below is the *superposition* of the two cases in (a) and (b). So what's V_{R2} ?



Original circuit

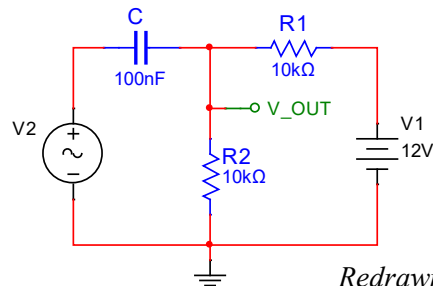
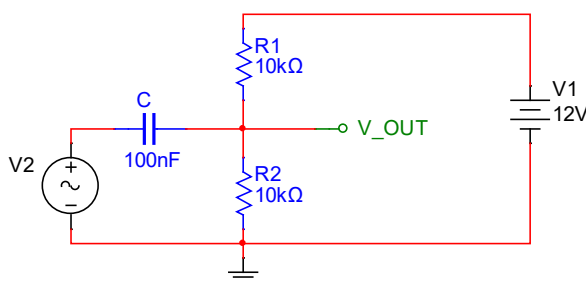


Shorting V_1 in part (a)



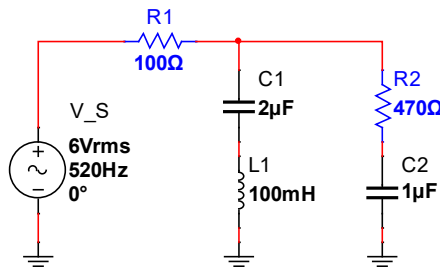
Shorting V_2 in part (b)

MC17. In this problem you will again analyze a circuit (below) using the *principle of superposition*, and those of you with a flair for observation will notice a certain similarity with the previous problem. (a) Imagine replacing the source V_2 with a short, and calculate the voltage V_{out} , assuming the circuit has been on long enough that the capacitor is fully charged. (b) With source V_2 restored, replace the source V_1 with a short. What kind of filter is the resulting circuit, and what is its cutoff frequency? With V_1 shorted out, what is V_{out} in the limit of both high and low frequencies? (c) The actual voltage V_{out} for the circuit as drawn below is the *superposition* of the two cases in (a) and (b). So what is V_{out} , in the limit of both high and low frequencies?

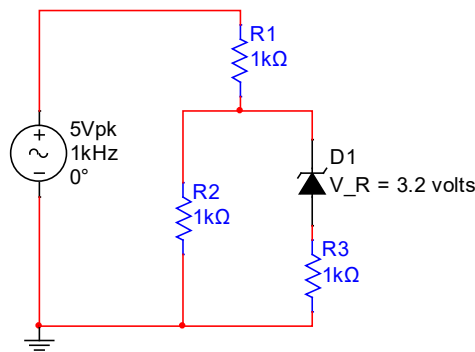


Redrawn to look exactly like MC16.

MC18: For the circuit below (from Paynter, page 448, problem 38), use Mathematica as a complex number calculator as we did in class to find the *magnitude* and *phase* of the voltage across capacitor C_2 .



MC19: In the circuit below, the Zener diode has $V_F = 0.7$ V and $V_R = 3.2$ volts. (a) Find the instantaneous current through resistor R_1 at the time of the peak positive source voltage (+5 volts). (b) Find the instantaneous current through resistor R_1 at the time of the peak negative source voltage (−5 volts). *Hint: One way to analyze the circuit in the case that you assume you know V_{diode} is to treat the diode as a known voltage source (a battery) and use the principle of superposition as you did in problem MC16.*

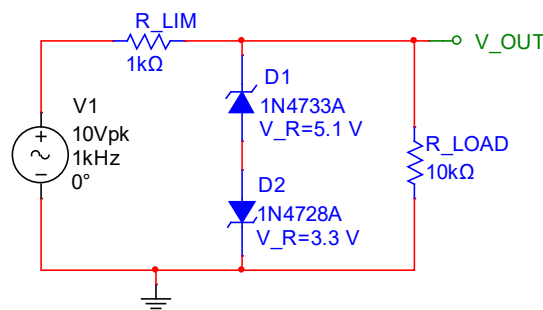


MC20: In problem MC13, you found that for a low-pass RC filter,

$$\frac{|V_{\text{out}}|}{|V_{\text{in}}|} = \frac{1}{\sqrt{(\omega RC)^2 + 1}}.$$

(a) Rewrite this expression in terms of the frequency f and the cutoff frequency f_c . (b) Suppose you have just designed a part of a circuit that gives a sinusoidal output signal at 100 Hz. Unfortunately, you realize that the output voltage also includes some interference at 25 kHz from a nearby video scanner circuit, which you decide to reduce by applying a low-pass RC filter to your output. What cutoff frequency f_c should you choose for your filter to reduce the interference by a factor of about 100, while still allowing at least 90% of your signal to pass?

MC21: The diagram below shows a clipper circuit with two Zener diodes. (a) Draw a graph of the output voltage vs. time, labeling the minimum and maximum values. (b) Calculate the maximum magnitude of the instantaneous current through the diodes. (c) Find the maximum instantaneous power dissipated in diode D2 and in the resistor R_{lim} .



Answers:

MC1. (b) 921 μsec (c) 250 μA

MC2. (a) 2.974 and 2.026 volts (c) 27.9 μsec (d) also 27.9 μsec (e) 17.92 kHz

MC3. (a) 144 Ω (b) 4.64 volts RMS (c) 150 mW (d) 6.12:1 (e) 240 mW

MC4. (a) 2.2 V (b) 2.8 V and 2.2 V (c) 2.8 mA and 5.5 mA (d) We can't have $I_{R2} > I_{R1}$. If we did, that would mean that *negative* current flows through the diode, that is, flowing in the reverse direction. That directly contradicts the assumption that the diode has a forward current. (e) open (f) $V_{R1} = 3.57 \text{ V}$; $V_{R2} = 1.43 \text{ V}$; $I_D = 0$.

MC5. $V_{R1} = 2.2 \text{ V}$; $V_{R2} = 3.1 \text{ V}$; $V_{R3} = 6.7 \text{ V}$; $I_{R1} = I_{R3} = 2.2 \text{ mA}$; $I_{R2} = 1.6 \text{ mA}$; $I_D = 0.6 \text{ mA}$

MC6. $V_{R1} = 1.33 \text{ V}$; $V_{R2} = 2.67 \text{ V}$; $V_{R3} = 4.0 \text{ V}$; $I_{R1} = I_{R2} = I_{R3} = 1.33 \text{ mA}$; $I_D = 0$

MC9. (b) I_{avg} (c) 19 mA (d) 0.55 Watts (e) 31.8 V

MC10. (a) 7 V (b) 2.52 V (c) 3300 μF (d) 10.92 V (e) maximum of 13.6:1

MC11. (a) Max value of R_{lim} is 800 Ω . (b) 938 μA (c) 20 mW and 9.38 mW, respectively

MC12. (a) 28.2 mA (b) Yes, they are always the same; otherwise components would build up static charges and start attracting each other with large Coulomb forces, which would be dangerous and embarrassing. (c) 2.68 V

MC13. (a) $|I| = V_0 / \sqrt{R^2 + 1/(\omega C)^2}$; $\phi = \tan^{-1} \frac{1}{\omega RC}$ (b) $|V_C| = V_0 / \sqrt{(\omega RC)^2 + 1}$; $\phi = -\tan^{-1} \omega RC$ (c) $f_C = 1/2\pi RC$ (d) about -3 dB (e) As $\omega \rightarrow 0$, $V_C/V_0 = 1$. As $\omega \rightarrow \infty$, $V_C/V_0 \approx (\omega RC)^{-1}$.

MC14. (a) 332.4 Ω *Hint: use your expression for $|I|$ in MC13 to find an expression for $|V_R|$.* (b) Zero. That's why you do this with a capacitor instead of a resistor. (c) 1.52 μF (d) No, Matt is a dumbass for buying the wrong bulbs in the first place.

MC15. (a) $|V_C| = V_0 / \sqrt{1 + R^2(1/\omega L - \omega C)^2}$; $\phi = \tan^{-1}[R(1/\omega L - \omega C)]$
(c) (i): $|V_{\text{out}}/V_{\text{in}}| \rightarrow \omega L/R$; $\phi \rightarrow +90^\circ$ (ii): $|V_{\text{out}}/V_{\text{in}}| \rightarrow 1/\omega RC$; $\phi \rightarrow -90^\circ$
(iii): $V_{\text{out}}/V_{\text{in}} = 1$; $\phi = 0^\circ$

MC16. (a) 3 V (b) 1.5 V (c) 4.5 V

MC17. (a) 6 V DC (b) High pass, $f_c = 318 \text{ Hz}$ (c) $f \gtrsim f_c$: $V_{\text{out}} = V_2 + 6 \text{ V}$; $f \ll f_c$: $V_{\text{out}} \approx 6 \text{ V}$

MC18. 3.719 V; -34.32° . (Is your answer for the magnitude off by a factor of $\sqrt{2}$? Remember that the "6 volts" on the source is an RMS value....)

MC19. (a) 2.5 mA (b) -3.1 mA. *Hint: if you got 2.27 mA for part (a), you probably assumed that there is current through the diode, so that the voltage across it is 3.2 V. But as you learned in problem MC4, it's also possible that there is no current through the diode. One of these scenarios leads to a physically plausible solution, and the other does not; you'll need to try both to see which one is correct.*

MC20. Anything between $f_c = 206 \text{ Hz}$ and $f_c = 250 \text{ Hz}$ will work fine.

MC21. (b) 5.6 mA (c) 18.5 mW and 36 mW.